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Emf_eca0502

Aim:

To write program for Transformation between Cartesian, Cylindrical and spherical coordinate system

using SCILAB.

The screenshot displays the SCILAB 10.0.0 environment. The main window shows a script named 'emf 1.0.sci' with the following code:

```
1 clear;
2 // Input Cartesian coordinates
3
4 // Input Cartesian coordinates
5 x = input('Enter the value of X: ');
6 y = input('Enter the value of Y: ');
7 z = input('Enter the value of Z: ');
8
9 // Cartesian to Cylindrical
10 // -----
11 r_cyl = sqrt(x^2 + y^2);
12 phi = atan2(y, x) * 180 / %pi; // Angle in degrees
13
14 disp('Cartesian to Cylindrical Coordinates: (r, phi, z) = ');
15 disp([r_cyl phi z]);
16
17 // Cartesian to Spherical
18 // -----
19 r_sph = sqrt(x^2 + y^2 + z^2);
20
21 if r_sph == 0 then
22     theta = 0;
23 else
24     theta = acos(z / r_sph) * 180 / %pi; // Polar angle in degrees
25 end
26 phi = atan2(y, x) * 180 / %pi; // Azimuth angle
27
28 disp('Cartesian to Spherical Coordinates: (r, theta, phi) = ');
29 disp([r_sph theta phi]);
30
```

The console window shows the execution results:

```
Enter the value of X = -1
Enter the value of Y = 3
Enter the value of Z = -5

"Cartesian to Cylindrical Coordinates (r, phi, z) = "
3.1622777 108.43495 -5.
"Cartesian to Spherical Coordinates (r, theta, phi) = "
5.9160798 147.68847 108.43495
--> |
```

The Variable Browser on the right shows the following variables:

Name	Value	Type	Visibility	Memory
phi	108	Double	local	216 B
r_cyl	3.16	Double	local	216 B
r_sph	5.92	Double	local	216 B
theta	148	Double	local	216 B
x	-1	Double	local	216 B
y	3	Double	local	216 B
z	-5	Double	local	216 B

The Command History window shows the following commands:

```
4
3
6*10^-2
6
1*10^-2
3*10^-2
2*10^-2
3
2
1
-1
-3
-5
```

The News feed window shows the following text:

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Given:

$$q = 3 \times 10^{-6}$$

$$r = 4 \times 10^{-3} \text{ m}$$

Formula

$$E = \frac{q}{4\pi\epsilon_0 r^2}$$

$$r^2 = (4 \times 10^{-3})^2 = 16 \times 10^{-6} = 1.6 \times 10^{-5} \text{ m}^2$$

$$4\pi\epsilon_0 r^2 = (4)(3.1416)(8.854 \times 10^{-12})(1.6 \times 10^{-5})$$
$$= 1.7803 \times 10^{-15}$$

$$E = \frac{3 \times 10^{-6}}{1.7803 \times 10^{-15}} = 1.685 \times 10^9 \text{ V/m}$$

$$E = 1.576 \times 10^9 \text{ V/m}$$

Electric Flux Density (D)

Formula:

$$D = \epsilon_0 E$$

$$D = (8.854 \times 10^{-12})(1.685 \times 10^9)$$

$$D = 1.492 \times 10^{-2} \text{ C/m}^2$$

$$D = 0.0013951$$

11.

Test case (ii) Transformation of Cylindrical coordinate points into Cartesian and spherical coordinate system

Scilab 2026.1.0 Console

File Edit Control Applications ?

Scilab 2026.1.0 Console

Enter the value of r = 3

Enter the value of phi (degrees) = 30

Enter the value of z = -4

"Cylindrical to Cartesian Coordinates (x, y, z) ="
2.5980762 1.5 -4.

"Cylindrical to Spherical Coordinates (r, theta, phi) ="
5. 143.1301 30.

-->

Variable Browser

Name	Value	Type	Visibility	Memory
phi	30	Double	local	216 B
phi_sph	30	Double	local	216 B
r	3	Double	local	216 B
r_sph	5	Double	local	216 B
theta	143	Double	local	216 B
x	2.6	Double	local	216 B
y	1.5	Double	local	216 B
z	-4	Double	local	216 B

Command History

```
// -- 27/07/2026 09:57:28 -- //
1
2
3
-1
-3
-5
3
30
-4
dc
3
30
-4
```

News feed

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emf 1.1.scilab (C:\Users\abhis\Documents\emf 1.1.scilab) - SciNotes

File Edit Format Options Window Execute ?

emf 1.1.scilab (C:\Users\abhis\Documents\emf 1.1.scilab) - SciNotes

emf 1.0.scilab emf 1.1.scilab

```
1 clear;
2 %clear;
3
4 // Input cylindrical coordinates
5 r = input("Enter the value of r: ");
6 phi = input("Enter the value of phi (degrees): ");
7 z = input("Enter the value of z: ");
8
9 //-----
10 // Cylindrical to Cartesian
11 //-----
12 x = r * cos(phi);
13 y = r * sin(phi);
14
15 disp("Cylindrical to Cartesian Coordinates (x, y, z) = ");
16 disp([x y z]);
17
18 //-----
19 // Cylindrical to Spherical
20 //-----
21 r_sph = sqrt(r^2 + z^2);
22
23 if r_sph == 0 then
24     theta = 0;
25 else
26     theta = asin(z / r_sph);
27 end
28
29 phi_sph = phi;
30
31 disp("Cylindrical to Spherical Coordinates (r, theta, phi) = ");
32 disp([r_sph theta phi_sph]);
33
```

Given Values:

$$\begin{aligned}r &= 3 \\ \phi &= 30^\circ \\ z &= -4\end{aligned}$$

Formula:

$$\begin{aligned}x &= r \cos \phi \\ y &= r \sin \phi \\ z &= z\end{aligned}$$

Calculation:

$$x = 3 \times \cos(30^\circ)$$

$$x = 3 \times 0.866 = 2.5981$$

$$y = 3 \times \sin(30^\circ)$$

$$y = 3 \times 0.5 = 1.5000$$

$$z = -4$$

$$(x, y, z) = (2.5981, 1.5000, -4)$$

Formula:

$$R = \sqrt{r^2 + z^2}$$

$$\theta = \cos^{-1}\left(\frac{z}{R}\right)$$

$$\phi = \phi$$

Calculation:

$$R = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

$$\theta = \cos^{-1}\left(\frac{-4}{5}\right) = \cos^{-1}(-0.8) = 143.13^\circ$$

$$\phi = 30^\circ$$

$$(R, \phi, \theta) = (5, 30^\circ, 143.13^\circ)$$

Test case (iii) Transformation of spherical coordinate points into Cylindrical and Cartesian coordinate system

The screenshot displays the Scilab 2026.1.0 environment. The main window shows a script file named 'emf 3.2.sci' with the following code:

```
1 // Clear workspace
2 clear;
3
4 // Input spherical coordinates
5 x = input('Enter the value of x: ');
6 phi = input('Enter the value of phi (degrees): ');
7 theta = input('Enter the value of theta (degrees): ');
8
9 // Spherical to Cartesian
10 //-----
11 r = x * sind(theta) * cos(phi);
12 y = x * sind(theta) * sind(phi);
13 z = x * cos(theta);
14 disp(['Spherical to Cartesian Coordinates (x, y, z) = ']);
15 disp([x y z]);
16
17 // Spherical to Cylindrical
18 //-----
19 r_cyl = x * sind(theta);
20 phi_cyl = phi;
21 z_cyl = x * cos(theta);
22 disp(['Spherical to Cylindrical Coordinates (r, phi, z) = ']);
23 disp([r_cyl phi_cyl z_cyl]);
24
```

The console window shows the execution results:

```
Enter the value of x = 10
Enter the value of phi (degrees) = 45
Enter the value of theta (degrees) = 130

"Spherical to Cartesian Coordinates (x, y, z) = "
5.4167522  5.4167522  -6.4278761
"Spherical to Cylindrical Coordinates (r, phi, z) = "
7.6604444  45.  -6.4278761
--> |
```

The Variable Browser window displays the following data:

Name	Value	Type	Visibility	Memory
phi	45	Double	local	216 B
phi_cyl	45	Double	local	216 B
r	10	Double	local	216 B
r_cyl	7.66	Double	local	216 B
theta	130	Double	local	216 B
x	5.42	Double	local	216 B
y	5.42	Double	local	216 B
z	-6.43	Double	local	216 B
z_cyl	-6.43	Double	local	216 B

The Command History window shows the following commands:

```
-- 27/07/2026 09:57:28 -- //
-1
-3
-5
-3
-30
-4
dc
3
-30
-4
10
45
130
```

The News feed window displays the following information:

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Calculation

$$r = 10 \times 0.7071 = 7.071$$

$$\phi = 130^\circ$$

$$z = 10 \times 0.7071 = 7.071$$

$$(r, \phi, z) = (7.071, 130^\circ, 7.071)$$

$$x, y, z = 7.66044, 45, -6.4278761$$

$$r = 10$$

$$\phi = 45^\circ$$

$$\theta = 130^\circ$$

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$

$$\sin 130^\circ = 0.766044$$

$$\cos 130^\circ = -0.6427876$$

$$\sin 45^\circ = 0.7071068$$

$$\cos 45^\circ = 0.7071068$$

$$x = 10 \times 0.766044 \times 0.7071068$$

$$x = 5.4167522$$

$$y = 10 \times 0.766044 \times 0.7071068$$

$$y = 5.4167522$$

$$z = 10 \times (-0.6427876)$$

$$z = -6.4278761$$

$$P(X, Y, Z) = (5.4167522, 5.4167522, -6.4278761)$$

Formula
 $p = r \sin \theta$
 $\phi = \phi$
 $z = r \cos \theta$

$$p = 10 \times 0.7660444$$

$$p = 7.660444$$

$$z = 10 \times (-0.642787876)$$

$$z = -6.4278781$$

$$P(p, \phi, z) = (7.6604444, 45^\circ, -6.4278761)$$

$$P(X, Y, Z) = (5.4167522, 5.4167522, -6.4278761)$$

$$P(p, \phi, z) = (7.6604444, 45^\circ, -6.4278761)$$

$$Q = 3 \times 10^{-6}$$

$$r = 4 \text{ mm}$$

Result:

Thus, the program was verified Transformation between Cartesian and Cylindrical coordinate system its

classified two categories (a) Cartesian to cylindrical and spherical coordinate system (b) Cylindrical to

cartesian and spherical coordinate system (c) spherical to cartesian and cylindrical coordinate system

using SCILAB was successfully.